



**CALICUT UNIVERSITY – FOUR-YEAR UNDERGRADUATE
PROGRAMME (CU-FYUGP)
GENERAL FOUNDATION COURSE
MULTIDISCIPLINARY COURSE (MDC)**

Programme	GENERAL FOUNDATION COURSE				
Course Code	ENG3FM107(2)				
Course Title	UNDERSTANDING KERALA: SCIENCE, CULTURE AND LITERATURE				
Type of Course	MULTIDISCIPLINARY COURSE (MDC)- SCIENCES				
Semester	3				
Academic Level	100-199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	3	3	-	-	45
Pre-requisites	Basic knowledge about Kerala's journey in science.				
Course Summary	The course aims at imparting the history and development of science in Kerala and how the contemporary socio-cultural scenarios of development are linked to science. It also provides glimpses of movements of scientific temperament, the modern institutions of scientific importance, technological developments and the representations of science media of Kerala.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO 1	To understand, and identify the history and legacy of science in Kerala	R, U	F, C	Debate, Role Play, Creative Writing, Article writing
CO 2	To familiarize with the interconnections of literary and other creative works in Kerala and how that has contributed to the scientific sensibility of the society.	U, An, Ap, C	C, P, M	Literary Review/Translation, Poster creation, Screening of various visual media presentations, Translation
CO 3	To foster a deeper understanding of the scientific temperament, working towards an egalitarian society.	U, An, E, C	C, P, M	Group projects, Panel discussions, Conduct interviews, Book reviews
CO 4	To develop ideas about the technological advancements of Kerala and its science institutions and how the learners can partake in the process.	Ap, An, C	C, P	Presentations, Industrial visits, Digital content creation
CO 5	To create sensibility on co-existence and to encourage practices in the maintenance of sustainable environment.	An, E	C, P, M	Newspaper reporting, Surveys, Conducting Interviews
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hours	Marks
I	Science in Kerala		9	10
	1	Excerpt from ‘Science in our Society’ – Dr B Iqbal https://www.youtube.com/watch?v=dzOT2pIn49E	3	
	2	Excerpt from ‘Science Dasakam’ – Sahodaran Ayyappan	2	
	3	An Unusual Route to Science- Minnie M Mathan from <i>Lilavati's Daughters: The Women Scientists of India</i> . Ed. Rohini Godbole & Ram Ramaswamy.	2	
	4	A Forgotten Episode in the History of Science Popularisation in Kerala- Urmila Unnikrishnan https://alablog.in/issues/54/history-of-science-popularisation/	2	
		Suggested Activities: 1. Conduct a debate on ‘Science and Gender’. 2. Role play of a scientist from Kerala 3. Write a short story about the life of a scientist from Kerala. 4. Write an article about a scientific contribution from Kerala.		

II	Kerala- Technology and Culture		9	15
	5	Kuttippuram Palam – Edasseri Govindan Nair (Trans. AsokakumarEdasseri& Jayasree) https://www.edasseri.org/English/translations/Kuttippuram%20Bridge%20by%20Edasseri%20-%20Translation.pdf	2	
	6	No Mere Ape- Excerpt from <i>The Tell-Tale Brain</i> - V. S. Ramachandran	3	
	7	<i>Android Kunhappan Version 5.25</i> – Dir. Ratheesh Balakrishnan Poduval https://app.primevideo.com/detail?gti=amzn1.dv.gti.12b78d12-d5b1-17dd-1c40-641f4fd88ed4&ref_=atv_lp_share_mv&r=web	3	
	8	An Introduction to Kerala State Council for Science, Technology and Environment https://kscste.kerala.gov.in/about-kscste/	1	
		Suggested Activities: 1. Identify and write a review of any literary work about the technology and culture of Kerala. 2. Organise a visit to one of the recognised science institutions from Kerala and create a short video of the same. 3. Conduct a survey on the Kerala-specific scientific scenario. 4. Write a book review on any Popular Science work.		

III	Kerala and Scientific Temper		9	15
	9	Excerpt from ‘Science for Social Revolution’- Dr. B. Ikbal & Dr. Thomas Isaac. <i>Science for Social Revolution: A Reader</i> . Ed. M. P. Parameswaran	2	
	10	Excerpt from ‘Understanding Scientific Temper’- Webinar- Vaisakhan Thampi https://www.youtube.com/watch?v=p9DQx5N_hx8	3	
	11	Papathara- Sarah Joseph <i>The Masculine of ‘Virgin’</i> : Stories of Sarah Joseph. Trans. J. Devika	2	
	12	Astronomy is not Astrology – Excerpt from a Conversation between Thushara GS Pillai and Sangeetha Chenampulli <i>Mathrubhumi Weekly</i> - 2021 March 14	2	
		Suggested Activities: 1. A group project analysing the superstitions prevalent in Kerala society. 2. Panel discussion on ‘Literature and Science’. 3. Interview with a Scientist/ Writer on Science. 4. Screen movies/ Documentaries relevant to the module.		

IV	Science and Kerala in Media		9	10
	13	A Living Fossil: The Purple Frog's Brief Life above Ground- <i>RoundGlass Sustain-</i> https://www.youtube.com/watch?v=x8flrb7WPOU	2	
	14	<i>Aavasavyuham</i> - Dir. Krishand https://www.sonyliv.com/movies/aavasavyuham-malayalam-1000182321	3	
	15	<i>The Birth</i> - Dir. Vimal Chandran https://www.youtube.com/watch?v=fAnwt-yclmk	2	
	16	India's First Digital University Comes up in Kerala- <i>Times of India</i> https://timesofindia.indiatimes.com/india/indias-first-digital-university-comes-up-in-kerala/articleshow/81126501.cms	2	
		Suggested Activities: 1. Conduct a poster exhibition on the unique fauna and flora of Kerala. 2. Identify themes and make presentations on science and the minority community in India. 3. Translate an article on science in Kerala. 4. Write a newspaper report on any technological development of Kerala.		
V	Open-Ended		9	

Note: The course is divided into five modules, with four having total 16 fixed units and one open-ended module with a variable number of units. There are total 36 transaction hours for the fixed modules and 9 hours for the open-ended one. Internal assessments (25 marks) are split between the open-ended module (5 marks) and the fixed modules (20 marks). The final exam, however, covers only the 16 units from the fixed modules.

Suggested References:

1. Flora- Kerala Forest and Wildlife Department
<https://forest.kerala.gov.in/index.php/about-us/2022-11-29-07-11-17/flora>
2. Kerala Declares Man-Animal Conflict a State-Specific Disaster: What Changes with This?-*The New Indian Express*
<https://indianexpress.com/article/explained/kerala-man-animal-conflict-state-specific-disaster-explained-9201527/>
3. Scientific Temper and Certain Popular Superstitions
<https://www.youtube.com/watch?v=G96mA9bg8ew>
4. Anatomy and Physiology of Science Capital: Kerala Model- Dr. Jayaraman A. P.
<https://www.youtube.com/watch?v=trsqz2k9Ka4>
5. In a First, ‘Queer Friendly’ Hospitals to be Introduced in 4 Kerala Districts- *India Today*
<https://www.indiatoday.in/india/story/in-a-first-kerala-to-implement-queer-friendly-hospital-initiative-2414606-2023-08-01>
6. Explore the Digital University Kerala, The Country's First Digital University
<https://www.youtube.com/watch?v=DXvSGaM511M>
7. 36th Kerala Science Congress- ‘Genesis’ & ‘Objectives and Scope’
<https://ksc.kerala.gov.in/genesis/>
<https://ksc.kerala.gov.in/objectives-and-scope/>
8. Science for Social Revolution: Science and Culture in Kerala- Govindan Parayil
Impact of Science in Society- UNESCO
9. Why Kerala Urgently Needs a Roadmap to Ecological Security- Pulapre Balakrishnan & Srikumar Chattopadhyay
<https://science.thewire.in/politics/government/why-kerala-urgently-needs-a-roadmap-to-ecological-security/>
10. Birds of Kerala- Salim Ali

Mapping of COs with PSOs and POs:

	PSO 1	PSO 2	PSO 3	PSO 4	PSO 5	PSO 6	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
C O 1	1	2	2	1	2	3	2	2	1	1	1	-	1
C O 2	3	3	3	3	2	3	3	1	3	2	-	1	3
C O 3	2	3	3	2	2	3	3	3	2	2	3	3	1
C O 4	3	2	2	3	2	3	2	3	3	3	3	2	3
C O 5	3	3	3	2	2	3	3	3	2	2	3	3	3

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

1. Quiz/ Discussion/ Seminar
2. Internal Exam
3. Assignment
4. End Semester Exam

Mapping of Cos to Assessment Rubrics:

	Quiz/ Discussion/ Seminar (10%)	Internal Exam (10%)	Assignment (10%)	End Semester Exam (70%)
CO 1	✓	✓	✓	✓
CO 2	✓	✓	✓	✓
CO 3	✓	✓	✓	✓
CO 4	✓	✓	✓	✓
CO 5	✓	✓	✓	✓